Before

Qwen3.8-27B

T Text

Image / video

Token embeddings

Vision encoder ×27

Projector

Embedding merge

64 layers

GDN

FFN

GDN

FFN

GDN

FFN

Dense attention

FFN

48 GDN + 16 Dense

×16

Dense attention

All historical KV retained

h

Read all past tokens

Q

Attention

E

Final norm

h

MTP ×1

Dense attention

FFN · shared LM head

Embed

LM head

248,320

Repeat

logits

Configurable drafts

Accept / reject

Output tokens

After

Qwen3.8-27B-QSA

T Text

Image / video

Token embeddings

Vision encoder ×27

Projector

Embedding merge

64 layers

GDN

FFN

GDN

FFN

GDN

FFN

QSA

FFN

48 GDN + 16 QSA

×16

QSA attention

All historical KV retained

h

Indexer

H4 · d128 · r4

Up to 512 blocks × 4

+ current block

Q

Attention

16 indexers · +52.4M FP32

E

Final norm

h

MTP ×1

Dense attention

FFN · shared LM head

Embed

LM head

248,320

Repeat

logits

Configurable drafts

Accept / reject

Output tokens

Base weights preserved

Added training

Custom native runtime · MTP on